COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	July 8-10, 2026	**DATE SUBMITTED:**	July 10, 2026
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Seatwork 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm:

Start the program
Declare the value of total score and average sum to 0
Declare the variables of Quiz 1, Quiz 2, Quiz 3
Input the score of each quiz
Add the score of each quizzes to compute for sum
Divide the total value of three quizzes by 3 to compute for the average
Output the result of total score and average score
End

Pseudocode:

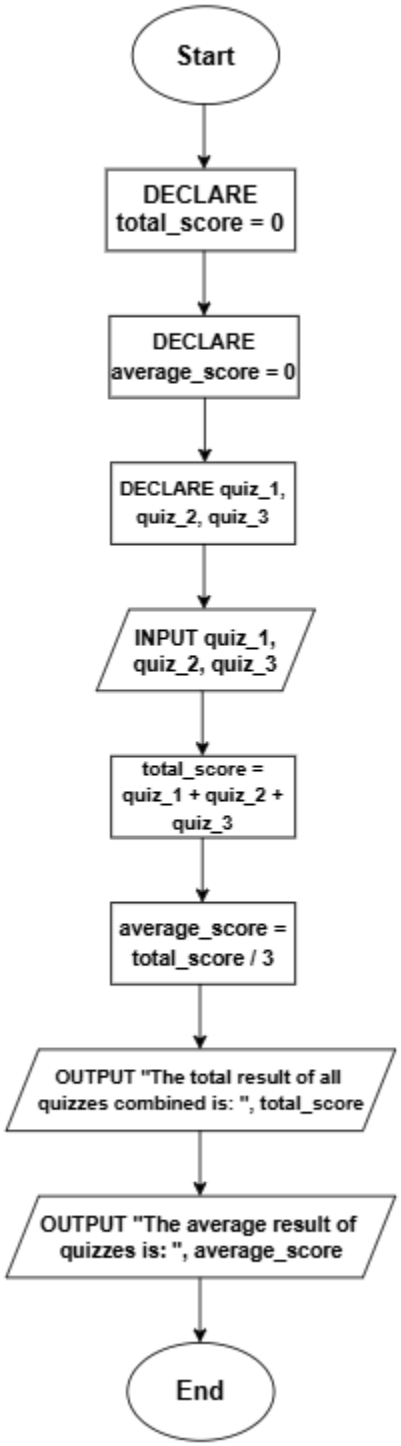
```
START
   DECLARE total_score = 0
   DECLARE average_score = 0
   DECLARE quiz_1, quiz_2, quiz_3

   INPUT quiz_1, quiz_2, quiz_3

   total_score = quiz_1 + quiz_2 + quiz_3
   average_score = total_score / 3

   OUTPUT "The total result of all quizzes combined is: ", total_score
   OUTPUT "The average result of quizzes is: ", average_score
END
```

Flowchart:


Start
DECLARE total_score = 0
DECLARE average_score = 0
DECLARE quiz_1, quiz_2, quiz_3
INPUT quiz_1, quiz_2, quiz_3
total_score = quiz_1 + quiz_2 + quiz_3
average_score = total_score / 3
OUTPUT "The total result of all quizzes combined is: ", total_score
OUTPUT "The average result of quizzes is: ", average_score
End

Analysis: For this program, I began by declaring the necessary variables for the process, specifically for the individual quiz scores, the total score, and the average score. After setting up the variables, I implemented the input process (cin) to capture the score for each individual quiz. Once the user enters these scores, the system adds them together to calculate the total combined score. After the total score is computed, the program divides it by three to determine the average. Finally, once the average score is resolved, the program outputs both the total and average scores to the screen.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

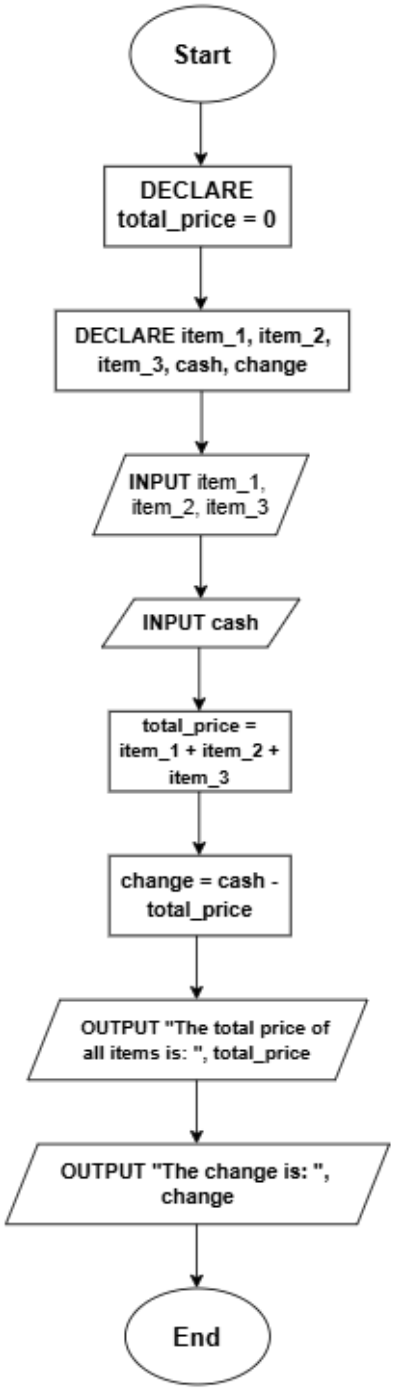
Algorithm:

Start the program
Declare the value of total price
Declare the variables of Item 1, Item 2, Item 3, cash, and change
Input the price of each items
Input the value of cash
Add the price of each items to compute for total price
Subtract the total price from cash to compute for change
Output the amount of total price and change
End

Pseudocode:

```
START
   DECLARE total_price = 0
   DECLARE item_1, item_2, item_3, cash, change
   INPUT item_1, item_2, item_3
   INPUT cash
   total_price = item_1 + item_2 + item_3
   change = cash - total_price
   OUTPUT "The total price of all items is: ", total_price
   OUTPUT "The change is: ", change
END
```

Flowchart:



Analysis: For the cashiering program, I followed a similar logic by first declaring the essential variables: the price of each item, the total purchase price, the cash rendered by the customer, and the resulting change. Next, I coded the input process to accept the price of each item, followed by the amount of cash provided. Once those values are received, the program sums up the individual

item prices to calculate the total bill. The system then subtracts this total price from the cash received to compute the exact change. Lastly, the program displays the total cost and the calculated change to complete the transaction.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Pseudocode Assistance, Grammar Refinement and Correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI to refine the format of pseudocode and seek some assistance for pseudocode part to accomplish it. Also, I used AI to refine my grammar and correct grammar mistakes to express my analysis properly and clearly.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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